

The Fulfillment Process

LEARNING OBJECTIVES

After completing this chapter you will be able to:

1. Describe the organizational levels associated with the fulfillment process.
2. List and explain the master data associated with the fulfillment process.
3. Identify the key steps in the fulfillment process and the data, documents, and information associated with each step.
4. Discuss the role of the credit management process in fulfillment.
5. Effectively use SAP® ERP to execute the key steps in the fulfillment process.
6. Explain how and why fulfillment is integrated with other processes.
7. Utilize SAP ERP to extract meaningful information about the fulfillment process.

GBI currently uses a very simple process to fill customer orders. This process, introduced in Chapter 1, is reproduced in Figure 5-1. The process begins when GBI receives a customer's purchase order, which it validates and authorizes via a sales order. The warehouse then prepares and sends the shipment, after which accounting forwards an invoice. The process ends when GBI receives payment from the customer.

This simple process has worked well for GBI up to this point. Because GBI has grown rapidly, however, the company wants to utilize the new ERP system it acquired to enhance its process for filling customer orders. Specifically, it wants to make the process more efficient and customer friendly as well as more transparent by keeping track of the status of every order throughout each step in the process. In addition, it wants to incorporate other sales-related activities, such as developing leads and responding to inquiries, into the new process.



Figure 5-1: A basic fulfillment process

In this chapter, we discuss the fulfillment process in detail, with a special focus on how an ERP system supports the process. We begin by identifying the key organizational levels and the master data related to the process. Next we examine the process steps in detail, and we explain how fulfillment is integrated with other processes. We conclude by discussing the various reports related to fulfillment. Unlike the previous chapter, this chapter does not include a separate section on key concepts. The reason for this is that the basic concepts associated with fulfillment, such as goods movement, have been addressed in previous chapters.

To illustrate the various concepts and process steps, we will use the following scenario throughout the chapter. Rocky Mountain Bikes (RMB), a GBI customer located in Denver, Colorado, has placed an order for 40 silver deluxe touring bikes and 100 t-shirts that it plans to sell at two racing events in Colorado Springs, one to be held on May 15 and the other on June 20. Because the bikes represent a significant inventory and storage expense, RMB wants GBI to deliver them directly to the racing location just a few days before each race. Further, RMB anticipates that the May race will attract a much larger crowd, so it expects to sell more bikes and t-shirts at that race. It has therefore requested that GBI deliver 30 of the bikes on May 10 and the remaining 10 on June 10. However, it wants GBI to deliver all 100 t-shirts in May so that it can sell as many as possible at the first race and then sell the remaining ones at the second race. Unlike the bikes, t-shirts are relatively inexpensive and are easy to store between races.

■ ORGANIZATIONAL DATA

Several organizational elements are essential to the fulfillment process. These are client, company code, sales area, plant, storage location, shipping point, and credit control area. Three of these—sales area, shipping point, and credit control area—are unique to fulfillment. A sales area is a combination of three other organizational elements—sales organization, distribution channel, and division—that are also unique to fulfillment. We discuss all of these organizational elements in this section.

Some elements—namely, client, company code, plant, and storage location—are also relevant to other processes and have been discussed in

previous chapters. In this section we have included a discussion of plant and storage locations as they relate specifically to fulfillment. No additional discussion of client and company code is necessary.

SALES ORGANIZATION

A company (company code) is divided into several **sales organizations**, each of which is responsible for the sale and distribution of goods and services for a particular geographical area, such as a regional or national market. Specifically, a sales organization is:

- Responsible for negotiating terms and conditions of sales for that market.
- Responsible to customers with regard to liability and rights of recourse in cases of disputes.
- The highest level of aggregation in sales-related reporting. That is, sales data can be summarized up to the level of the sales organization.

A company code must have at least one sales organization, although it can have many. The latter arrangement is appropriate if the sales processes are substantially different in the different sales organizations, for instance, to handle regional differences in practices and customs. In other cases a company might use multiple sales organizations simply to make sure that the geographic area covered remains manageable. Although a company can include multiple sales organizations, a sales organization can belong to only one company code. The sales organizations for GBI are depicted in Figure 5-2. As the figure indicates, GBI US has two sales organizations, one each for the Eastern and Western US. GBI Germany also has two sales organizations, one each for the Northern and Southern territories.

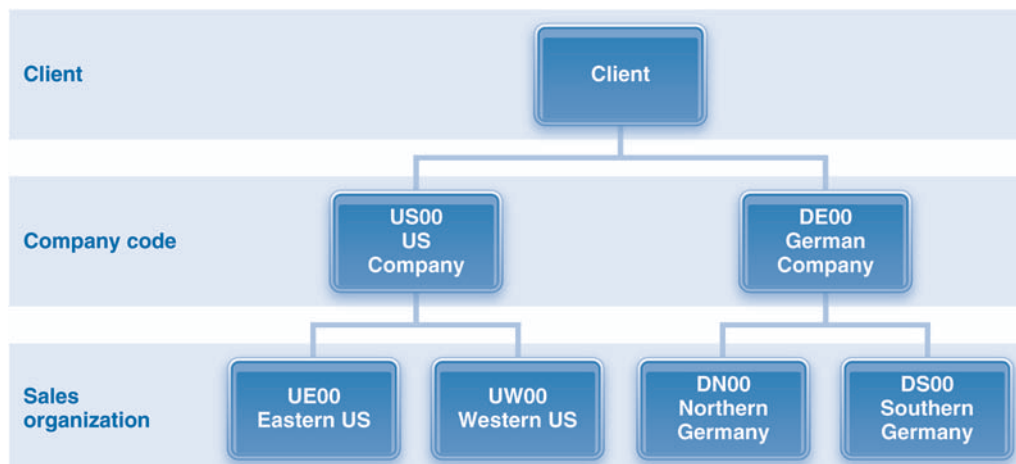


Figure 5-2: GBI sales organizations

Business Processes in Practice 5.1: Intel's Organizational Structure

Intel Corporation has six independent operating groups that manufacture products, each of which operates four sales organizations: Asia-Pacific, the Americas, Europe, and Japan. Intel locates its sales organizations close to their largest clusters of customers. Each of the 24 sales

organizations has two distribution channels, direct and reseller. Thus, there are 48 combinations through which Intel sells its products to its customers globally.

Source: Intel company reports.

DISTRIBUTION CHANNEL

A **distribution channel** (DC) is the means by which a company delivers its goods and services to its customers. Typical channels are wholesale, retail, and online (Internet sales). Just as a company can have multiple sales organizations, it can also have multiple DCs. Each channel has its own strategies, approaches, and constraints for getting the goods and services to the customer. More specifically, each channel has its distinctive responsibilities, pricing systems, plants from which shipments are made, and other characteristics. For example, a wholesale channel has the following characteristics (among others):

- It does not include sales taxes in calculating prices (in the United States).
- It requires a minimum volume of purchase and offers volume discounts.
- It may designate a specific plant or plants from which deliveries are made.

In addition, reporting can be consolidated at the DC level. That is, statistics can be summarized and aggregated based on distribution channels. A sales organization must have at least one distribution channel, although it can have more than one. In addition, a distribution channel can be assigned to multiple sales organizations.

Because GBI is a manufacturing organization, it historically has sold its products through the wholesale channel. Its customers are retailers who, in turn, sell the products to the end-consumers. Recently, however, GBI has begun to sell directly to end-customers via the Internet. Although anyone can access the GBI website to purchase a bike, factors such as complex taxes, shipping costs, and import duties make it very difficult for GBI to operate a global Internet sales channel. Therefore, GBI manages both the wholesale and Internet sales channels together at the country level, as illustrated in Figure 5-3. This policy ensures that the sales organization that ships the product to the customer is the group that is also the most familiar with the tax laws for that region.

Note that, although all four sales organizations identified in Figure 5-3 are involved with wholesale sales, only two are involved with Internet sales. The Western US sales organization manages Internet sales for the entire United States, and the Northern German sales organization manages Internet sales for all of Germany. If GBI were to venture into retail sales at a later date, it would create a new channel—retail—to manage these sales because the strategies for retail sales, such as pricing, minimum quantities, and taxes, will be different from the other two channels.

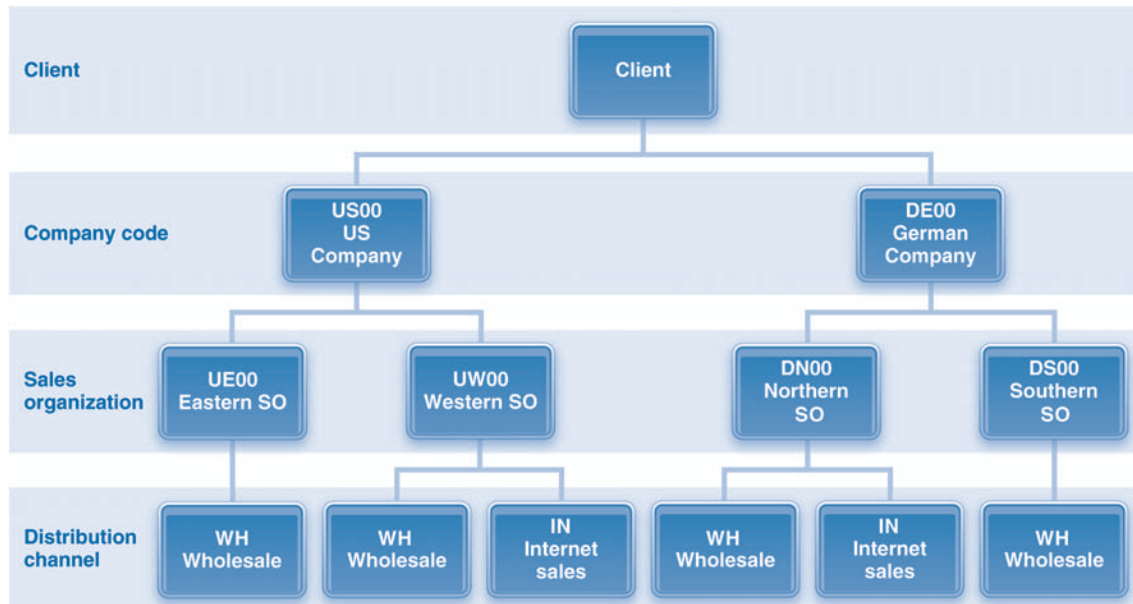


Figure 5-3: GBI US distribution channels

A unique combination of a sales organization and distribution channel is called a **distribution chain**. Some master data, such as material master and pricing conditions, are maintained at the distribution chain level.

Business Processes in Practice 5.2: Distribution Channels at Apple

Apple Inc. provides a good example of a company that utilizes multiple distribution channels. Apple sells its products through its online store (Internet), its Apple stores (retail), and third-party retailers (wholesale). Apple uses each of these channels to resolve different issues within the fulfillment process.

Because Apple places a priority on the customer's buying experience, it tries to establish either an Apple-owned retail store or a mass-market retail store in every major market. This policy ensures that the majority of consumers in most developed nations live within a short distance of a retail store that sells Apple products.

Although retail stores offer the best opportunity for consumers to examine the products before purchasing them, maintaining an extensive network of physical stores entails many major expenses, including real estate, local taxes, and retail staff. Therefore, Apple very carefully evaluates new locations for its stores based largely on the local demographics of those consumers who are the most likely buyers.

When it is not feasible for Apple to establish a retail store in an area with a strong potential consumer base, the company must partner with another retailer to resell its products. Although this arrangement eliminates many of the retail costs identified above for Apple, it also reduces Apple's profit per unit, because Apple must now share its sales revenues with the reseller.

Online retailing eliminates a great deal of the costs associated with operating physical stores. In addition, because the online channel sells directly to consumers, Apple does not have to share its profits with a reseller. However, because online retailers sell to any number of locations around the world, they must deal with shipping, taxes, and import costs that the retail channel does not have to. In addition, they must invest in the technical infrastructure that supports a global online retail site. Apple's online retail operations, including the Apple store and iTunes, are based on SAP ERP.

Source: Apple company reports.

DIVISION

Most companies consolidate materials and services with similar characteristics within a unit known as a **division**. Typically, each division is associated with a company’s product line. A product or material can be assigned to one division only. Each division can employ its own sales strategies, such as pricing agreements with customers. In addition, it is possible to aggregate reports at the division level.

A sales organization must have at least one division. A division can be assigned to multiple sales organizations, and a sales organization can include many divisions.

GBI currently has two divisions, the bicycles division and the accessories division. Recall from Chapter 2 that GBI’s material list includes different material types. The finished goods—that is, bicycles—are managed by the bicycles division, while the trading goods—such as helmets—are managed by the accessories division. In the future, if GBI expands into other product lines, such as skateboards, it will create additional divisions. The divisions for GBI US and GBI DE are illustrated in Figure 5-4 and Figure 5-5, respectively.

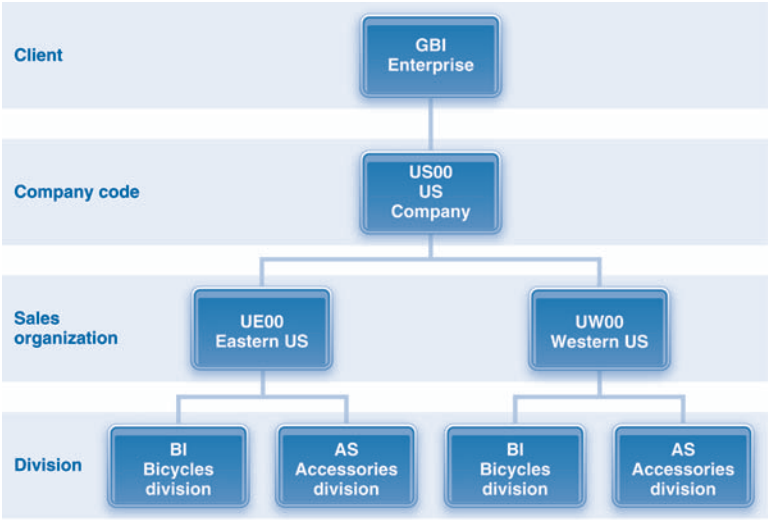


Figure 5-4: GBI US divisions

Business Processes in Practice 5.3: Multiple Divisions in a Major Publishing Company

John Wiley & Sons, the publisher of this textbook, has three main divisions that serve different reading markets around the world. Wiley publishes college textbooks through the Higher Education division. The Professional/Trade division contains the For Dummies, Frommer’s, Betty Crocker, CliffsNotes, and Webster’s Dictionaries series, as well as several other popular brands. Wiley’s third division is the Scientific, Technical, Medical, and Scholarly (STMS) division, which is the world’s largest publisher of professional and scholarly

journals and books. Although the physical products (books/journals) and the geographic regions are similar for all three divisions, the audience, authors, and distribution channels for each division vary greatly. Wiley has learned that by grouping its brands and the employees who are responsible for their success into three divisions, they can achieve greater synergies and operate more efficiently.

Source: John Wiley & Sons company reports.

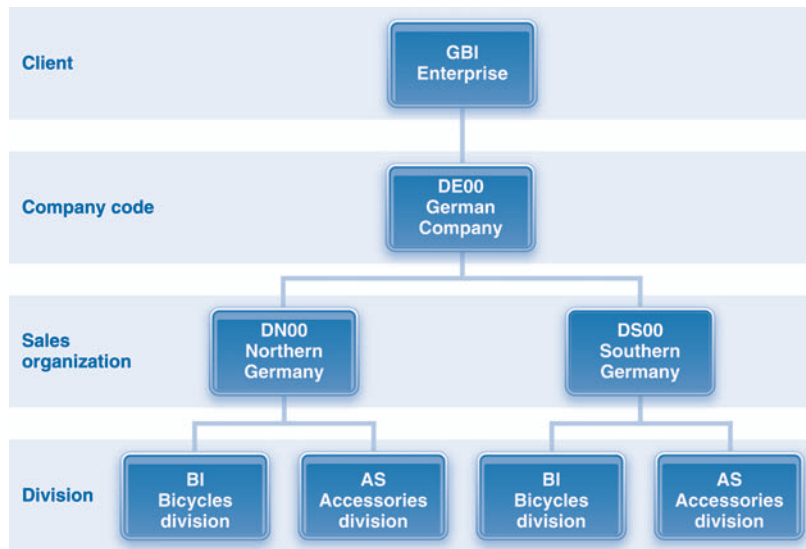


Figure 5-5: GBI DE divisions

SALES AREA

A **sales area** is a unique combination of sales organization, distribution channel, and division. In other words, it defines which DC a sales organization uses to sell the products associated with a particular division. A sales area can be assigned to only one company code. All of the documents associated with the fulfillment process, such as quotations and packing lists, belong to one sales area.

Figure 5-6 illustrates the six sales areas for GBI US. For example, one sales area (UE00+WH+BI) is responsible for the sale of products in the bicycles *division* for the Eastern US *sales organization* via the wholesale *channel*. This sales area is highlighted in the figure. A second sales area (UW00+WH+AS) manages the sale of products in the accessories division for the Western US sales organization via the wholesale channel. As a third example, sales area UW00+IN+BI manages the sale of products in the bicycles division for the

Business Processes in Practice 5.4: The Sales Structure at Apple

As we discussed in a previous example, Apple sells its products through a combination of online stores, Apple retail stores, and reseller stores. Apple contains several hardware divisions, including Mac computers, iPods, iPhones, servers, and accessories. Apple also operates several sales areas for different types of customers, such as education, government, enterprise, and consumer. Each of these customer segments has unique needs and requires a distinctive sales approach, even though they are buying the same products. Therefore, Apple must

plan its sales strategies by product globally while paying close attention to the unique characteristics of both the channels (retail, online, reseller) and the customer (education, government, enterprise, and consumer). This strategy is typically reflected in a matrix, which allows Apple to pinpoint opportunities and manage their global sales efforts effectively.

Source: Apple company reports.

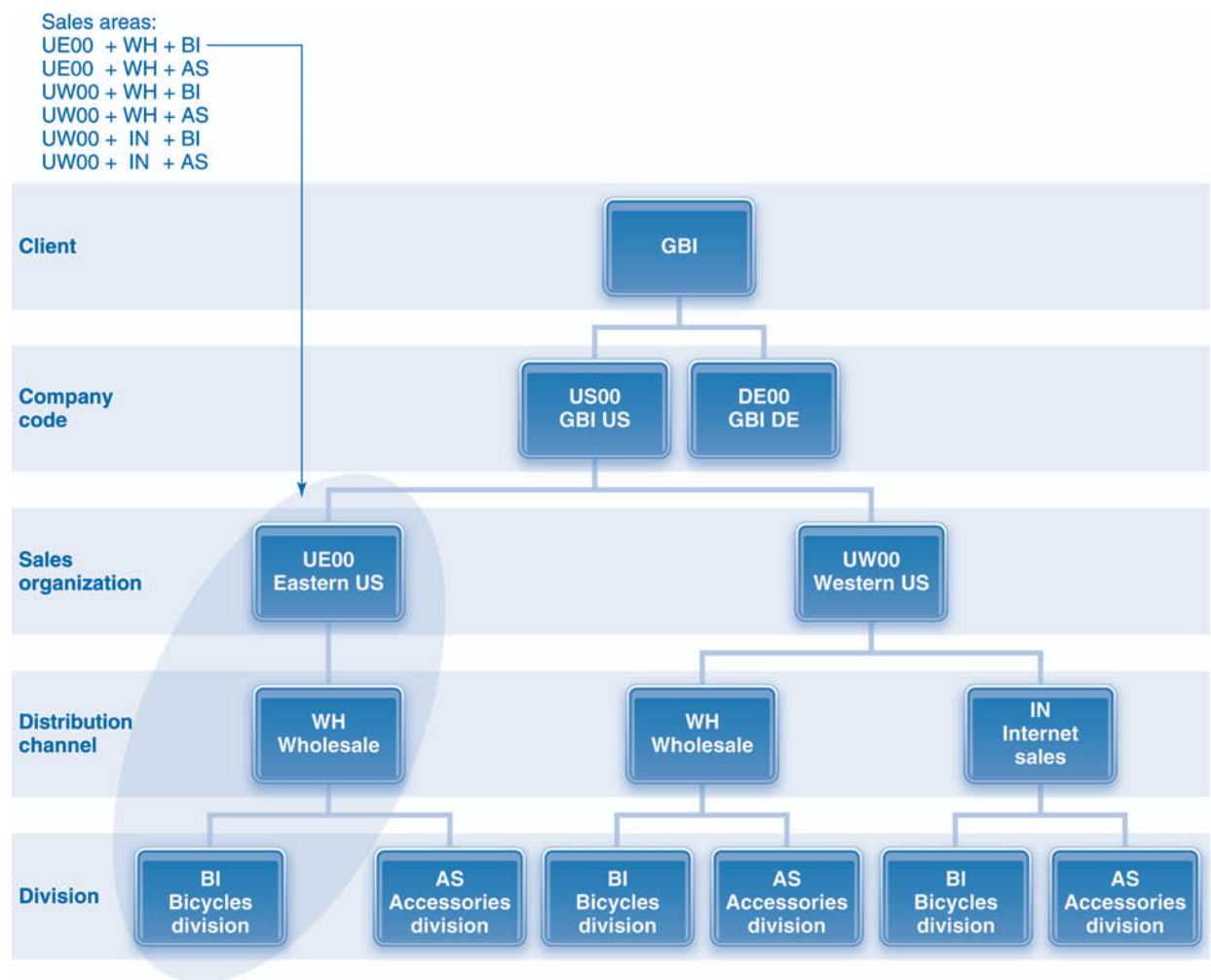


Figure 5-6: GBI sales areas

Western US sales organization via the Internet channel. The German company also has six sales areas. They are not depicted in the figure, however.

PLANT

We have discussed plants in previous chapters in the context of other processes. In the fulfillment process a *delivering plant* is a facility from which the company delivers products and services to its customers. In the case of products, a plant is typically a manufacturing and/or storage facility. In the case of services, it can simply be an office. A plant can be assigned to more than one distribution chain. Recall that a distribution chain is a unique combination of sales organization and distribution channel. Conversely, a distribution chain can be associated with more than one plant.

In Figure 5-7, plant 1 delivers only for distribution chain 1 and plant 3 only for distribution chain 3. In contrast, plant 2 delivers for all three distribution

chains. Viewed from a different perspective, distribution chains 1 and 3 use two plants to deliver materials to their customers, whereas distribution chain 2 uses only one plant.

Recall from Chapter 2 that GBI has three plants in the United States and two in Germany. In the United States, the Dallas plant is the sole manufacturing

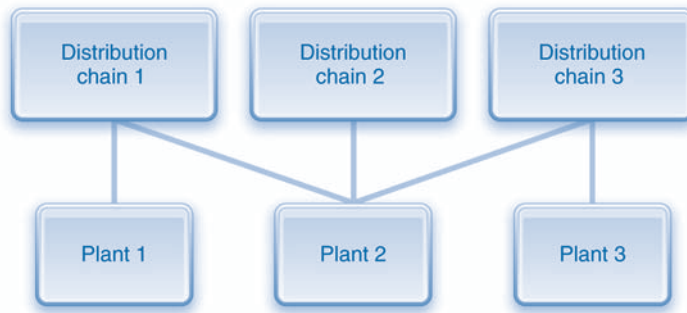


Figure 5-7: Plants

site, while the plants in San Diego and Miami serve as distribution centers. Finished goods are shipped from the Dallas facility to the other two facilities as needed. In contrast, trading goods are shipped directly to the two regional distribution centers by the vendors. The Dallas plant does not hold an inventory of trading goods. Normally, the Miami plant delivers products to customers in the Eastern United States and the San Diego to the Western United States. However, the Dallas plant serves as an overflow facility to the two other plants and can ship finished goods anywhere in the country, as needed. In Germany, the Heidelberg plant ships to locations in the South and the Hamburg plant to locations in the North.

SHIPPING POINT

A **shipping point** is a location in a plant from which outbound deliveries are shipped. It can be a physical location, such as a loading dock, a rail depot, or a mail room. It can also be a designated group of employees who, for example, handle express or special deliveries. A shipping point is associated with one or more plants, and a plant can have more than one shipping point. A plant must have at least one shipping point from which to process deliveries, although the shipping point does not have to be physically located within the plant.

Figure 5-8 diagrams a scenario in which multiple plants access one shipping point. The figure represents a campus for a company that includes a factory and two storage facilities, all of which are plants. Note that the only plant that has a shipping point is the storage facility located by the front office. Therefore, all materials from the factory and the main storage facility must first be moved to this facility and then shipped to their destinations.

In contrast, Figure 5-9 illustrates a scenario in which several plants share three shipping points. Storage facility #1 has one shipping point, and facility #3 has two. Storage facility #2 has none, so it uses the shipping points located in the

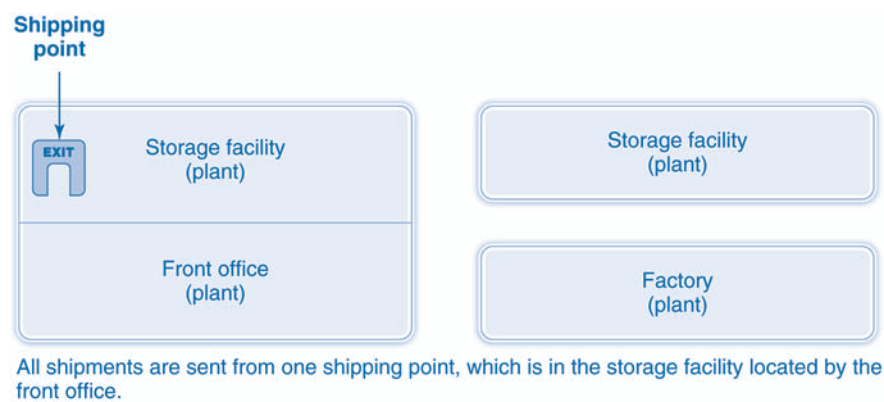
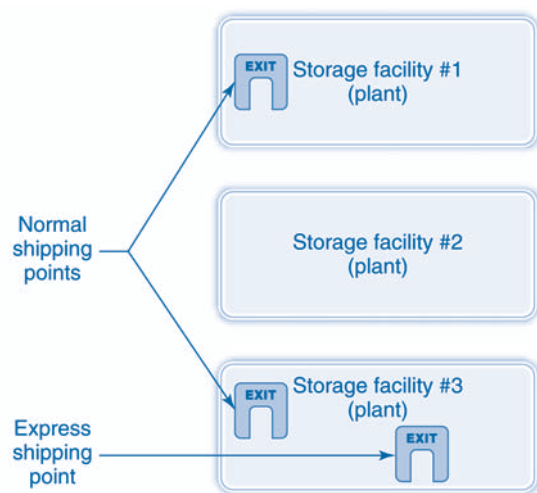


Figure 5-8: Shared shipping point



The express shipping point is used by all plants, as needed. Facility 2 uses the shipping points located at the other two facilities.

Figure 5-9: Multiple shipping points

other plants. In storage facility #3, one of the shipping points handles normal deliveries, while the other one is reserved for express deliveries or for deliveries that require special handling.

In GBI, each of the five plants has a single shipping point. All customer deliveries are shipped from one of these locations. Each shipping point is a loading dock, which is a facility that trucks can back into so they can be loaded. Figure 5-10 illustrates a loading dock. Other facilities that can operate as shipping points include rail depots (Figure 5-11) and shipyards or ports (Figure 5-12). In the case of a rail depot, the rail cars are pulled up to the facility and loaded. When the storage facility is located by a waterway, a shipyard serves the same purpose.



Figure 5-10: Loading dock as a shipping point. ©iStockphoto



Figure 5-11: Rail depot as a shipping point. ©iStockphoto



Figure 5-12: Port as a shipping point. ©iStockphoto

CREDIT CONTROL AREA

A **credit control area** is an organizational level that is responsible for customer credit. Specifically, it determines customers' creditworthiness, establishes credit limits, and monitors and manages the actual extension of credit to customers. An enterprise can choose to manage credit in either a centralized or a decentralized manner. In a centralized system, a single credit control area manages credit for customers across all company codes in the enterprise. This arrangement is particularly useful if customers conduct business with multiple company codes within the enterprise. As an example, Figure 5-13 illustrates a centralized model that GBI employs for customers who purchase bicycles and accessories from both GBI US and GBI DE. In such cases, GBI manages the

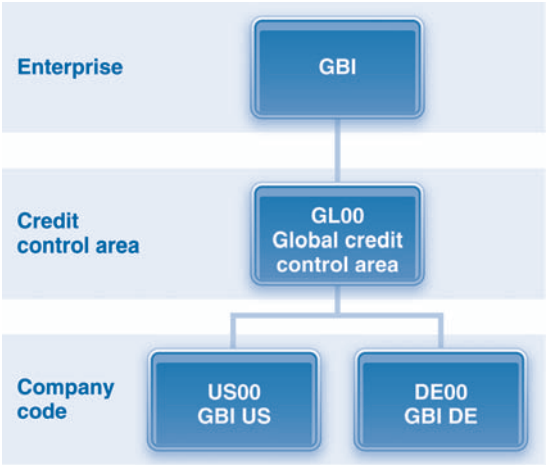


Figure 5-13: Centralized credit control area

credit limit at the enterprise level, meaning that it assigns all company codes to one credit control area.

In contrast, an enterprise that utilizes a decentralized model maintains multiple credit control areas, each of which manages credit for one or more companies within the enterprise. For example, in the hypothetical scenario illustrated in Figure 5-14, GBI has two credit control areas, one for all companies in North America (GBI US, GBI Canada, and GBI Mexico) and one for all companies in Europe (GBI DE and GBI Great Britain). If GBI were to expand operations into the Asia-Pacific region, it likely would create a third credit control area to supervise credit for customers of those companies. Even when using a decentralized approach, however, it is possible—and prudent—to establish credit limits for customers at the enterprise level and for each credit control area. This policy ensures that customers purchasing from companies belonging to different credit control areas (e.g., GBI US and GBI DE in Figure 5-14) do not exceed credit limits across the enterprise.

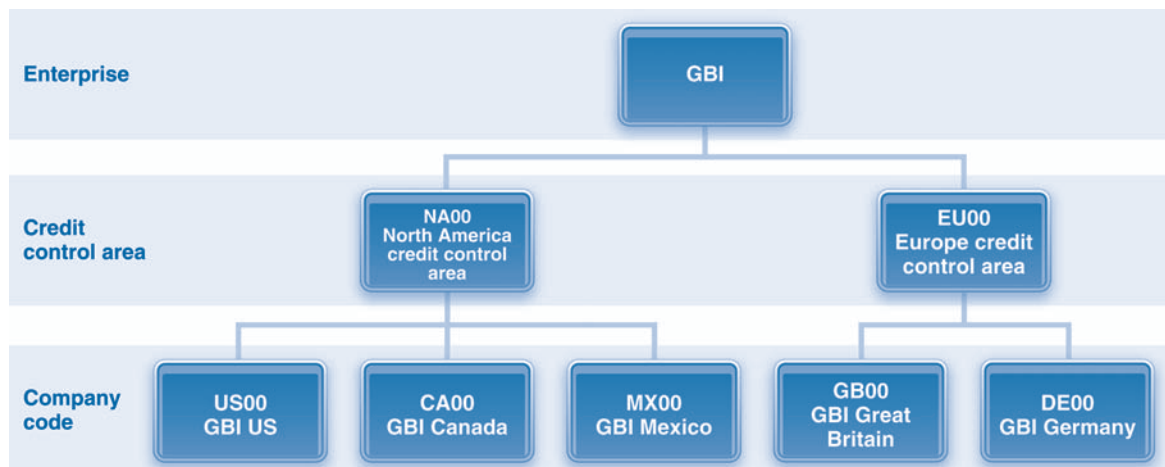


Figure 5-14: Decentralized credit control areas

MASTER DATA

In Chapter 4 we discussed the material master, pricing conditions, and output conditions from the perspective of procurement. In this section we will explore these master data as they pertain to fulfillment. In addition, we will discuss master data that are relevant only to fulfillment, namely, customer master, customer-material information record, and credit management master record.

MATERIAL MASTER

The key organizational elements in fulfillment for which material master data are defined are client, sales organization, distribution channel, and plant. Recall that material master data are grouped into views and that each view is relevant to one or more processes and defined for specific organizational levels. The three views relevant to fulfillment are basic data, sales organization data, and sales plant data.

Basic data, which are relevant to all processes, are defined at the client level. **Sales organization data** are defined for combinations of sales organizations and distribution channels. Examples of sales organization data are the delivering plant, sales units, and minimum quantities. The delivering plant is the preferred plant from which deliveries are made for the particular sales organization and distribution channel. Sales units are the units of measure, such as cartons, barrels, containers, cases, pallets, and crates, in which the materials are sold. Quantities include minimum order quantities and delivery quantities. In addition, a link to pricing conditions (discussed later) is available. Because the data are defined for each combination of sales organization and DC, each DC can have different values for these data to support different sales strategies. **Sales plant data** provide details on how the material will be shipped from that plant. Examples are specific transportation requirements (e.g., refrigeration) and the methods of loading the material (e.g., a hand cart, forklift, or crane). Note that it is necessary to define materials for every combination of sales organization and distribution channel, as well as for every plant if the data are different. Very often, the material data will be unchanged between plants or distribution channels.

Demo 5.1: Review material master

CUSTOMER MASTER

Customer master data include data needed to conduct business with customers and to execute transactions that are specifically related to the fulfillment process. The data in the customer master are divided into three segments—general data, accounting data, and sales area data. The relationships among these three segments and the two departments in an organization responsible for these data (accounting and sales) are depicted in Figure 5-15.

General data are defined at the client level. They are valid for all of a client's sales areas and company codes. Examples of general data are a customer's name, address, and account number. **Accounting data** are specific to

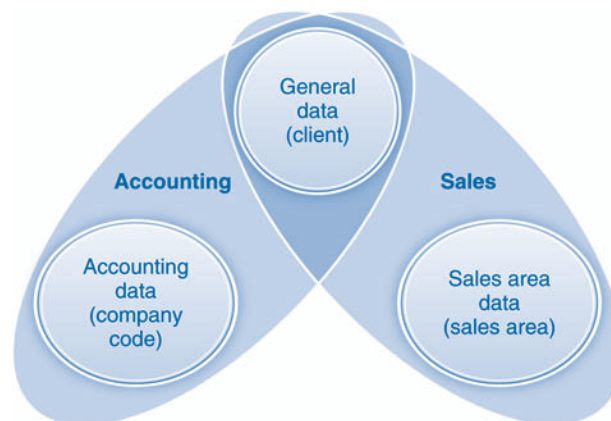


Figure 5-15: Segments of customer master data

a company code and include data such as payment terms and the reconciliation account in the general ledger. Recall from Chapter 3 that (1) customer accounts are subledger accounts, (2) subledger accounts are linked to the general ledger via reconciliation accounts, and (3) the accounts receivable account is typically used as the reconciliation account for customers. The accounting department will typically complete this part of the customer master data. **Sales area data** are specific to a particular sales area, which, as we discussed earlier in this chapter, is made up of one sales organization, one distribution channel, and one division. Sales area data relate to sales, shipping, billing, and partner functions. Examples are the sales office and the currency in which the transaction is conducted. Shipping data specify the preferred delivering plant, priorities, and methods. They also define delivery tolerances and policies for dealing with partial deliveries. Billing data include billing terms and tax-related data.

If a customer is served by multiple sales areas, then the data must be defined separately for each sales area. This arrangement permits a company to apply different terms and conditions for different areas. Figure 5-16 illustrates this approach for a hypothetical customer with offices in both the United States and Germany. The customer purchases materials from both GBI's US and Germany companies. The general data for the customer apply across both GBI companies—that is, the GBI enterprise that is represented by the client. However, accounting data for the customer must be defined separately for each GBI company. Recall that one piece of data in the customer master record is the reconciliation account, which is based on the chart of accounts. If GBI US and GBI DE use different charts of accounts, then they may have different reconciliation account numbers. Further, the currencies and tax-related data are also different. Thus, the accounting data are different for each country (company code).

Finally, the customer purchases materials for its facilities throughout the United States. Consequently, it deals with both US sales organizations (UE00 and UW00). In addition, the customer purchases exclusively on a wholesale basis. Therefore, master data for this customer must be defined for two sales areas in the United States UE00+WH+BI and UW00+WH+BI (see Figure 5-16).

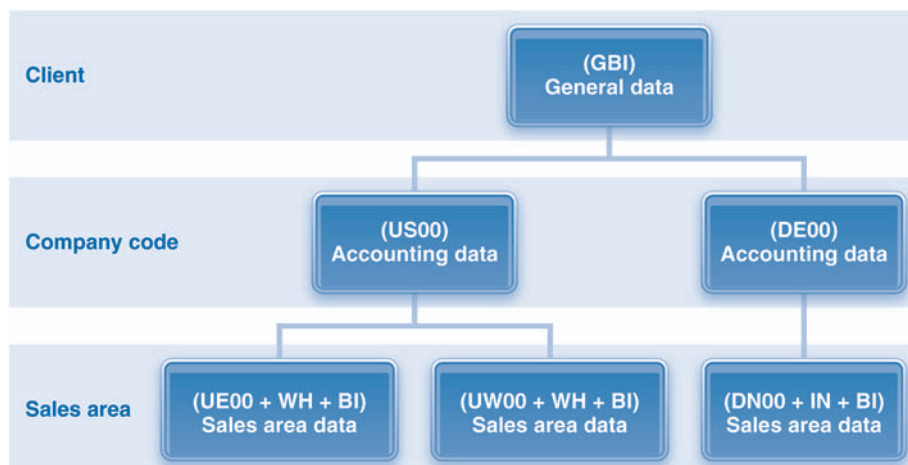


Figure 5-16: Multiple definitions of a customer

In contrast, the customer in Germany deals only in the northern part of the country and purchases exclusively via the Internet. Therefore, only one set of sales area data is defined for that customer.

Customers can play different roles or **partner functions** in the fulfillment process. The four required partner functions are *sold-to party*, *ship-to party*, *bill-to party*, and *payer*. One customer may fill all four roles, or each role may be filled by a different customer. The customer who submits the order is the **sold-to party**. This is the primary type of business partner. The order may indicate that the materials should be shipped to a different location or that the invoice should be sent to someone else. These are the **ship-to party** and **bill-to party** functions, respectively. For example, the customer may request that the material be shipped to one of *its* customers. In this case, the party that actually receives the shipment is considered the ship-to party. However, the invoice will be forwarded to the customer that ordered the materials—that is, the bill-to party. Alternatively, the bill-to-party may be another company that processes invoices for the customer. As another example, a large enterprise may be comprised of a number of companies, such as franchises. An order may be placed by headquarters, the sold-to party, with instructions to deliver the materials to numerous franchises, the ship-to parties. The invoice may be sent to headquarters, which also serves as the bill-to party. Finally, the payment may be outsourced to another company, which will be the **payer** of the invoice.

GBI US has 12 customers, and GBI DE has 7. These customers are listed in Figure 5-17. Additional details of the master data for each customer are provided in Appendix 5A. Rocky Mountain Bikes (RMB), the customer used in our example, is located in Denver, Colorado, and therefore is serviced through GBI's Western US sales organization. In addition, RMB purchases via the wholesale channel. Therefore, it is associated with the sales area made up of the Western US sales organization (UW00), the wholesale channel of distribution (WH), and the bicycle division (BI) (look back to Figure 5-6). Further, because RMB is a small company, its partner functions are usually all located at the same address. However, it is not uncommon for RMB to request that materials be shipped directly to various racing events or to its retail customers. In our example, RMB has requested that the materials be delivered to the RMB booth at the racing location in Colorado Springs, not to its head office in Denver. In this case, RMB is the sold-to party, bill-to party, and payer, but the ship-to party is the racing location. Finally, because the materials are to be delivered in Colorado, they will be delivered from the San Diego plant.



Figure 5-17: GBI customers

Demo 5.2: Review customer master

CUSTOMER-MATERIAL INFORMATION (INFO) RECORD

A **customer-material information record** is comprised of master data specific to one customer and one material. In contrast to data in a material master, which apply to all customers, and data in a customer master, which apply to all purchases made by a particular customer, data in a customer-material info record relate to purchases of a *specific material* by a *specific customer*.

Some data in a customer-material info record are not found elsewhere. One example is the *customer material number*, which cross-references the company's material number with the customer's material numbers. Figure 5-18 provides some examples of material numbers used by GBI and its customer RMB. RMB's material numbers are noted as the customer material number in the info record. This cross-referencing of material numbers enables the customer to place an order based on its internal material numbers, which the info record then translates into the company's material numbers. Remember that most customers also have enterprise systems that manage their procurement process. Consequently, the customer material number is the link between the seller's master data and the buyer's master data.

Material	GBI Material number	RMB Material number
Deluxe touring bike-black	DXTR1000	G1000BL
Deluxe touring bike-silver	DXTR2000	G1000SL
Deluxe touring bike-red	DXTR3000	G1000RD

Figure 5-18: RMB and GBI material numbers

In certain cases the data in a customer-material info record supersede data found in other master data, such as the material master and customer master. For example, preferences related to shipping, such as delivering plant, tolerances, and partial deliveries that are included in the customer master apply to all materials purchased by the customer. However, if these preferences vary for different materials, then they are included in the customer-material info record. For example, if deliveries are normally sent to one specified plant but deliveries for a particular material must be sent to a different plant, then this preference is noted in the customer-material info record for that material, not in the customer master.

The customer-material info records for RMB also note that RMB prefers the bikes to be shipped via special ground freight (truck) due to the weight and size of the shipping boxes. In contrast, the company requests that GBI ship the shirts via standard air freight (FedEx or UPS 2-day). Finally, the info records indicate that RMB will accept partial deliveries for shirts but not for bikes, because shirts have a very short lead time and can be reordered quickly if needed.

Demo 5.3: Review customer–material info record

PRICING CONDITIONS

Pricing conditions are master data that companies use to determine the selling prices of their products. Companies create conditions for various components of the final selling price, including *gross prices*, *discounts*, *freight*, *surcharges*, and *taxes*. Conditions can be fixed amounts, percentages, or based on a sliding scale, and they can be either independent of or relative to other conditions. For example, the price of a product can be material specific and customer independent, meaning that the seller charges the same price to all of its customers. Alternatively, the price can be customer specific, in which case the company charges different customers different prices, perhaps based on some agreements between the company and the customer. Similarly, discounts can be uniform, or they can be based on the quantity or value of the purchase. For example, GBI offers a 10% discount for purchases of between 100 and 500 units and a 20% discount for purchases of more than 500 units. Freight is generally based on the weight of the shipment, and it may be waived for purchases greater than a predetermined amount. Thus, the final price to the customer is a function of, or *conditioned* on, numerous variables.

Because numerous conditions are defined for a product, a company must have a procedure to determine which conditions apply to a particular customer order. This procedure, called the *condition technique*, consists of identifying available *condition types* (gross price, customer-specific price, discounts, freight, surcharges, etc.) and determining which ones apply to the particular circumstances of the order.

Recall that in our example, RMB wishes to purchase 40 bikes and 100 t-shirts. The bikes do not qualify for a discount, so GBI will charge the standard price of \$2,800 per bicycle. However, the t-shirts qualify for a 10% discount, so the price will be \$27 per shirt, reduced from the standard price of \$30.

Demo 5.4: Review pricing conditions

OUTPUT CONDITIONS

A variety of outputs that are generated during the fulfillment process, such as quotations, confirmations, and invoices, must be communicated to customers. The data needed to perform this task are included in **output conditions**. The condition technique used to determine pricing is also used to determine how outputs from the process are communicated to the customer.

Output conditions are defined separately for the different output types (quotations, invoices, etc.). Data in the condition master include the output medium (e.g., print, fax, EDI), partner function (e.g., sold-to party, ship-to party), and transmission time (e.g., immediately, or periodically using a program).

RMB prefers to receive order confirmations via e-mail within 4 hours of acceptance or shipment of an order. It also prefers to receive paper invoices. GBI must maintain all of these output conditions in its ERP system.

CREDIT MANAGEMENT MASTER RECORD

The **credit management master record** is an extension of the customer master record that includes data relevant to managing credit for that customer. The data contained in this record are grouped into three segments: general data, credit control area data, and an overview. The *general data segment* includes data applicable at the client level, that is, across multiple credit control areas. Examples of general data are address, communication data, and total credit granted to the customer across the enterprise. The *credit control area segment* includes data applicable to a single credit control area. An example is the credit granted to the customer for companies in a particular credit control area and risk category. *Risk category* is a classification used to determine how risky it is to extend credit to the customer. Companies use this classification to assess the likelihood that the customer will pay its invoices. Finally, the *overview segment* includes key data from the other segments from the credit master record. Companies use the overview segment to access the most important data they need to make decisions regarding extending credit to customers.

■ PROCESS

Figure 5-19 illustrates the steps in the fulfillment process. The process begins with presales activities and concludes with the receipt of payment from the customer. *Presales activities* are optional and are designed to identify and develop customer relationships. Very often, the process begins with *sales order processing*, which is triggered by the receipt of a customer's purchase order. As the name suggests, sales order processing involves creating a *sales order*, which is an internal document used to manage and track the order as it flows through the process. After a company creates the sales order, it prepares a *shipment* and sends it to the customer. The next step is *billing* the customer for the materials shipped. Finally, the company receives a *payment* from the customer. As you can see in Figure 5-19, each step involves several tasks. Further, depending on the circumstances, additional steps can be necessary, including activities to be completed by other processes in the organization. Note that in this example, we have not explicitly included credit management. We will assume that the customer has sufficient credit and that the order will be processed. We will, however, consider credit management activities and their impact in greater detail later in this chapter.

A variety of documents are created during the fulfillment process. As previously discussed, documents are categorized as *material documents*, *financial accounting (FI) documents*, *management accounting (CO) documents*, and *transaction documents*. Recall that documents can be printed or can exist in electronic form. The data contained in these documents are categorized as organizational data, master data, and transaction data. Some data are provided by the user, while other data are automatically retrieved from the relevant master data and the existing transaction documents.

This overview of the fulfillment process is highly simplified. The following discussion highlights each process step in detail in terms of its key elements: *triggers*, *data*, *tasks*, and *outcomes*.

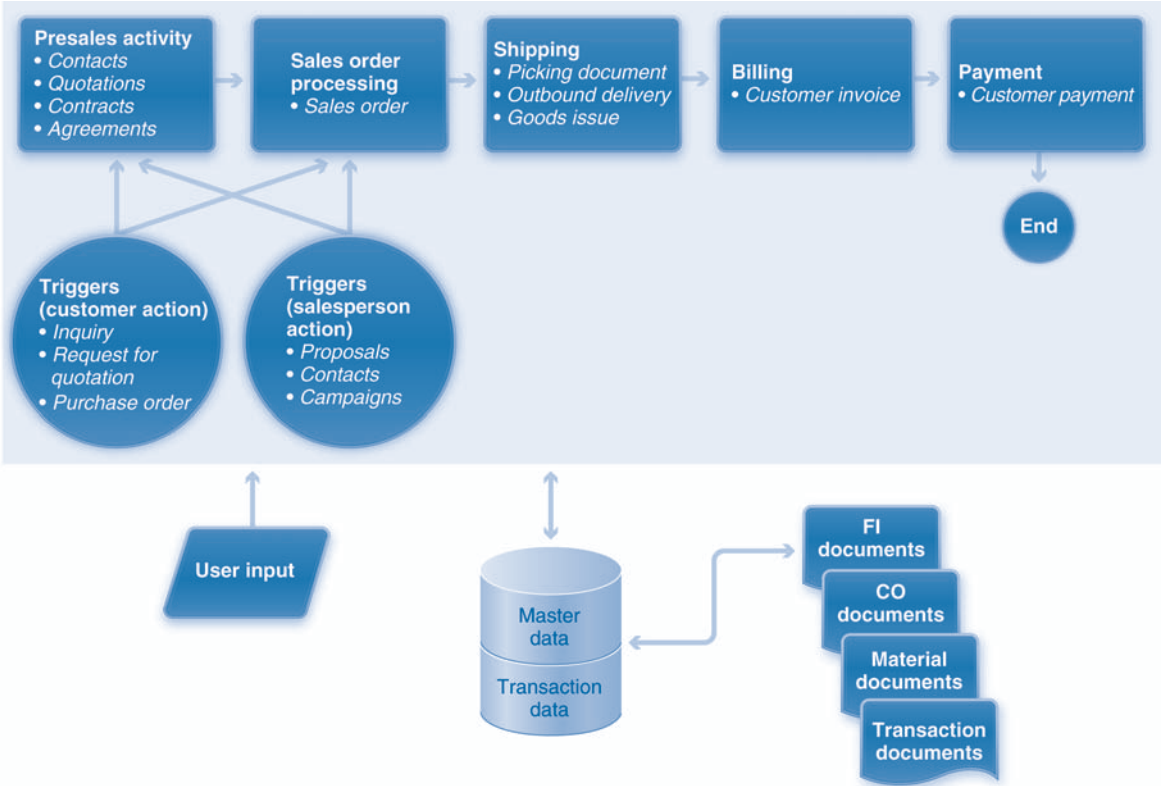


Figure 5-19: Fulfillment process steps

PRESALES ACTIVITY

Presales activity is often triggered by a communication from a customer such as an inquiry or a request for quotation (RFQ). The outcome is a quotation that is sent to the customer making the inquiry. The key elements of this inquiry–quotation activity are illustrated in Figure 5-20. An **inquiry** is a request for information regarding a potential order that the customer might place with the company. In our example, RMB inquires whether GBI can deliver 40 silver deluxe touring bikes and 100 t-shirts by the specified dates. If so, then RMB requests information on pricing, shipping costs, and discounts. In response, GBI creates a quotation and forwards it to RMB. A **quotation** is a binding agreement to sell the customer specific products under clearly defined delivery and pricing terms. Because these terms can change, a quotation will typically include a validity date, that is, a date up to which the quotation remains in effect.

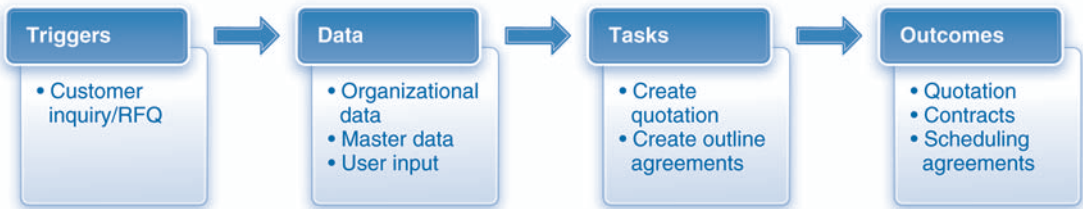


Figure 5-20: Elements of the presales activity

Besides sending quotations in response to inquiries, presales activities can include managing customer contacts and creating outline agreements (explained in the next paragraph). SAP ERP provides some capabilities to track data pertaining to *established* customers, such as their preferences and purchasing history. Companies can analyze these data to create marketing and sales strategies designed to encourage customers to place additional orders. In addition, the system can track *potential* customers. ERP systems typically manage basic presales activities. Companies utilize customer relationship management (CRM) systems, discussed in Chapter 2, to manage very detailed presales, sales-pipeline, sales prospects, and marketing activities that generate a quotation in the ERP system. The handoff between the presales processes in a CRM system and the quotation-to-cash process in the ERP system is a critical integration point for most companies because failures, such as missed sales opportunities, can significantly reduce sales revenues.

Just as quotations are binding agreements made by sellers, **outline agreements** are binding agreements made by *customers* to purchase specific quantities or values of materials. An example is a GBI customer who enters into an outline agreement to purchase 1,000 standard touring bicycles over a six-month period for a specified price. Alternatively, the agreement may specify value instead of quantity. In this case, the customer agrees to purchase bicycles valued at \$30,000 over the next six months. These types of outline agreements are called **contracts**. Another form of outline agreement includes specific delivery schedules. For example, a GBI customer could agree to purchase 1,200 bicycles over the next six months and to accept delivery of these bicycles on a specific schedule. In this case, the agreement is called a **scheduling agreement**.

Data

Several types of organizational and master data are necessary to process an inquiry and create a quotation. These data are presented in Figure 5-21. Particularly important are data concerning the customer, the materials, and pricing. User input consists of the customer number, material numbers, quantities, and dates. The SAP ERP system uses these inputs to obtain the necessary organizational data, such as sales area, from the various master data in the system. Recall that the material master and customer master are associated with specific organizational elements. SAP ERP also uses the customer number to obtain necessary customer data, such as contact information, from the customer master. Finally, the system uses material numbers to obtain availability and pricing data from available pricing conditions or customer-material info records.

The preceding paragraph deals with existing customers. If the inquiry is from a new customer, then the system will not contain either master data or a customer-material info record for that customer. In such cases, the company must first create the customer master data in the system. The info record is not essential because pricing data are available via pricing conditions.

The quotation can be created with or without reference to existing documents. If the quotation is created without reference, then all of the necessary data must be provided when the quotation is created. However, if a reference document—such as a previously created inquiry, quotation, sales order, or an outline agreement—is used, then data from these documents are automatically included in the quotation. These data are then updated as needed before the quotation is completed. Figure 5-22 diagrams the process of incorporating existing data into the quotation.